

Jessica Oldov

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EDUCATION

Duke University

Bachelor of Science in Computer Science, Minor in Russian Culture and Language

Aug. 2021 – May 2025

GPA: 4.00/4.00

Honors: QuestBridge Scholar, Allan Kornberg Scholarship Recipient

EXPERIENCE

Software Engineer

Liberty Mutual Insurance

July 2025 – Present

New York, NY

- Led the implementation of Teen-Driver Discount logic deployed to production in June 2026 for 3 states, processing **10K+** eligible policies/month and reducing qualifying premiums by **8%**
- Developed and implemented a **Java 21/Spring Boot microservice** integrating 120+ actuarial risk variables across **Kafka, MongoDB, and REST APIs**, reducing pricing variance by **7%** and improving annual premium accuracy by **\$12M+**
- Built and maintained automated test suites using **JUnit 5, Mockito, and Spring Boot Test**, catching critical pricing defects pre-deployment and maintaining **98%** coverage across core rating logic

Software Engineering Fellow

MLH Fellowship – Meta (PyTorch)

September 2024 – December 2024

Remote

- Optimized PyTorch model checkpoint serialization using **Python and C++**, refactoring tensor storage allocation to achieve a **15% reduction in serialization time** for large-scale models (1B+ parameters)
- Led sprint planning and issue triage for a 4-person contributor team, merging **12 PRs** into the PyTorch main branch and accelerating critical bug resolution by **25%**

Software Engineer Intern

Liberty Mutual Insurance

June 2024 – August 2024

Indianapolis, IN

- Developed a **TypeScript/Node.js serverless payment workflow** using **AWS Lambda, API Gateway, and DynamoDB**, automating **500+ daily transactions** and reducing manual processing time by **20%**
- Built conversational payment flows using **Amazon Lex and AWS Lambda** with telephony integration, deflecting **35%** of payment calls from live agents

Undergraduate Teaching Assistant – Computer Science

Duke University

January 2023 – May 2025

Durham, NC

- Mentored **150+ students** across 4 semesters in software engineering fundamentals (**Ruby on Rails, Sinatra**), consistently rated **among top TAs** by faculty and student course evaluations

PROJECTS

Small Claims Insurance Application | *Python, AWS Lambda, DynamoDB, Datadog, OpenAI API* August 2025

- Architected and built a **Python/AWS serverless claims platform** integrating the **OpenAI API** for conversational claim intake, increasing completion rates and reducing submission time from **12 to 4 minutes**
- Deployed the platform to **200+ users** within its first month and reduced API response latency by **30%** by optimizing **DynamoDB query patterns** and mitigating **AWS Lambda cold starts**

myLearning App | *React, TypeScript, Node.js, Express.js, Next.js, MongoDB, Redis*

July 2025

- Co-led a team of 4 to build a full-stack resource-sharing platform with **RESTful APIs, MongoDB aggregation pipelines** for keyword search, and Redis caching that reduced query response time by **45%**
- Won 3rd place** out of **40+ teams** at Liberty Mutual's annual Hackathon for advanced filtering and a gamified engagement system (XP, leaderboards)

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C++, SQL

Frameworks & Libraries: Spring Boot, React, Node.js, Next.js, Express.js

Cloud & Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, SQS), Kafka, Docker, PostgreSQL, MongoDB, Redis, Git, CI/CD

Data & ML: OpenAI API, PyTorch, TensorFlow, scikit-learn, pandas, NumPy